

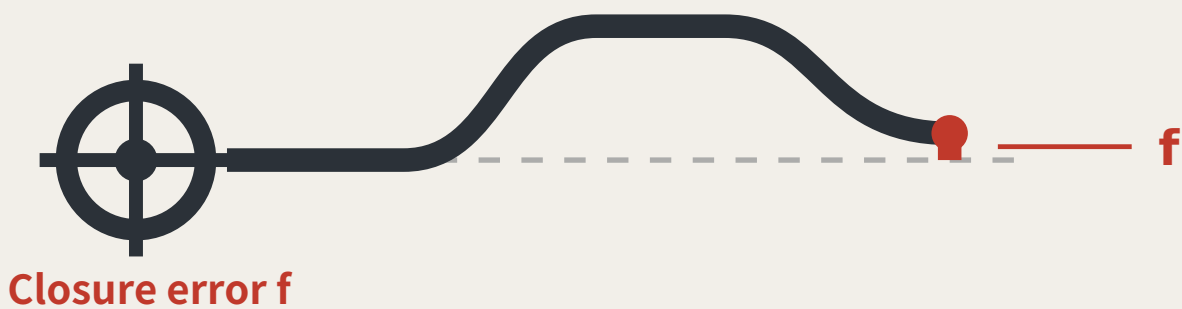
THE LEVELING LINE

A3 BOOKLET

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f .
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Two readings of the same point, and the trust metric for this corridor.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²

Overall design area

13.8 km/km²

Network density, measured

9,443 m

Spine length

17

Oversized blocks, not parks

8

Tiered benchmarks

29 / 33

Gates proven able to fail

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

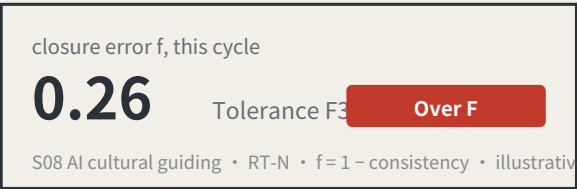
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

the L3 return-to-datum point

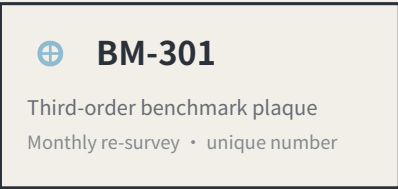
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

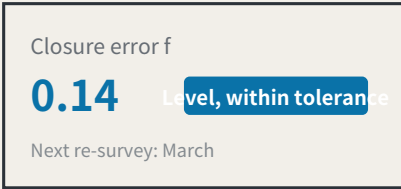
Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



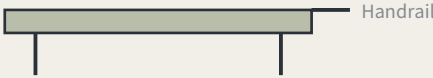
1 Stone and plaque

The number is the position, the order and the cycle Visible on site; no internet needed



2 Reading board

Visible on site; no internet needed



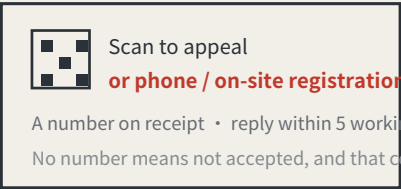
3 Dwellable seating

With armrests: older people rely on them to stand



4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	2nd-order point	Quarterly re-survey
BM-3xx	3rd-order point	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	third order BM-301/BM-302/BM-303 • P5 • P7
Quarterly	Scenario open day	second order BM-21/BM-22 P2 • P3
Semi-annual	Route re-survey	first order, routes RT-N / RT-S P1 • P2
Annual	Zeroing ceremony	the L3 return-to-datum point all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Id	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	led-high	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Act Two: turn the same instrument on the site

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
-------	--

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

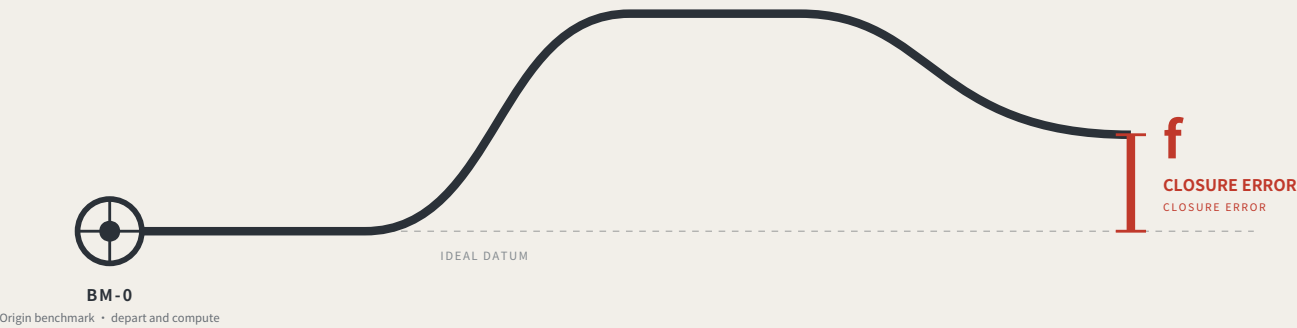
Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

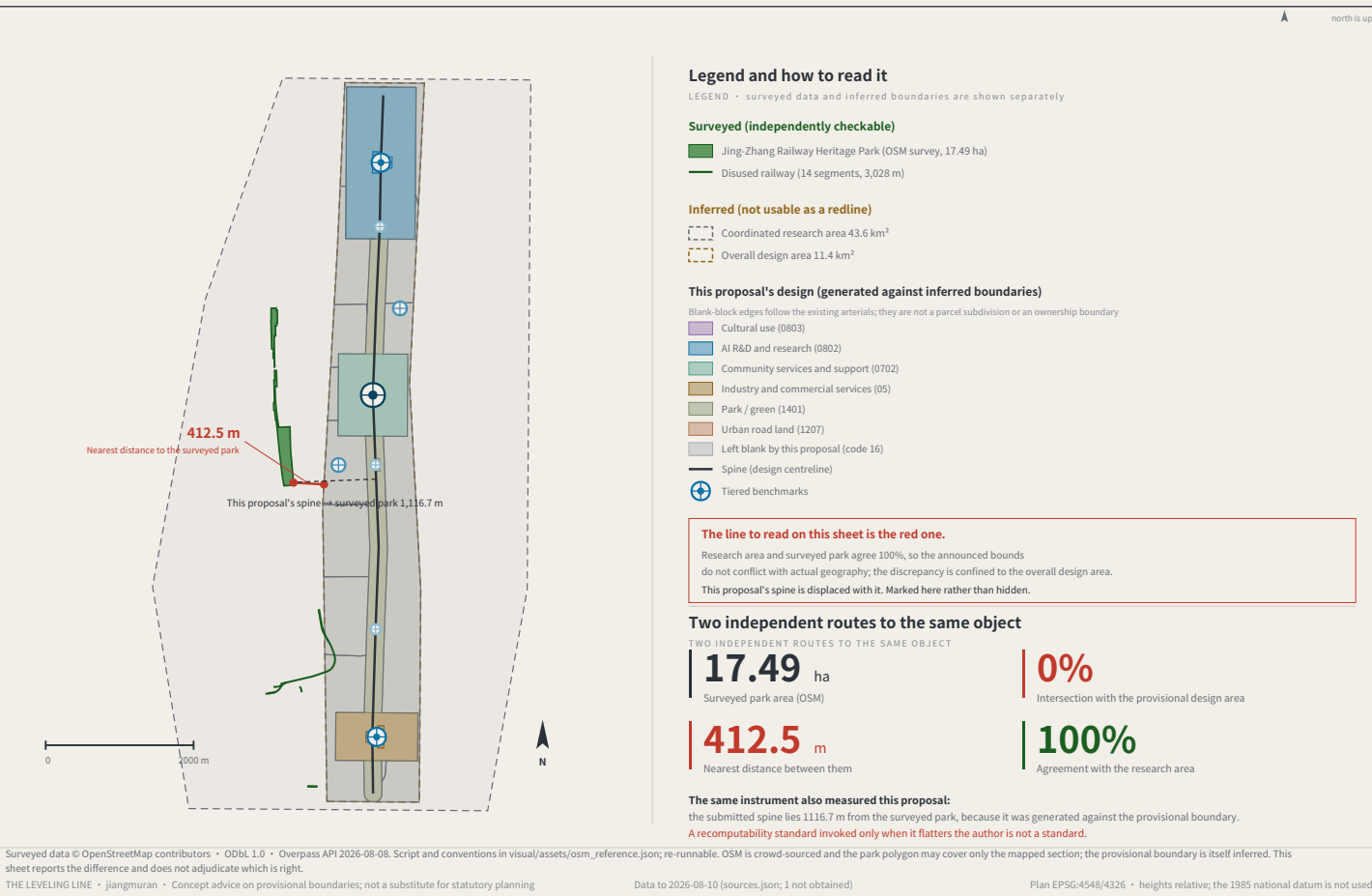
This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

FIG.00

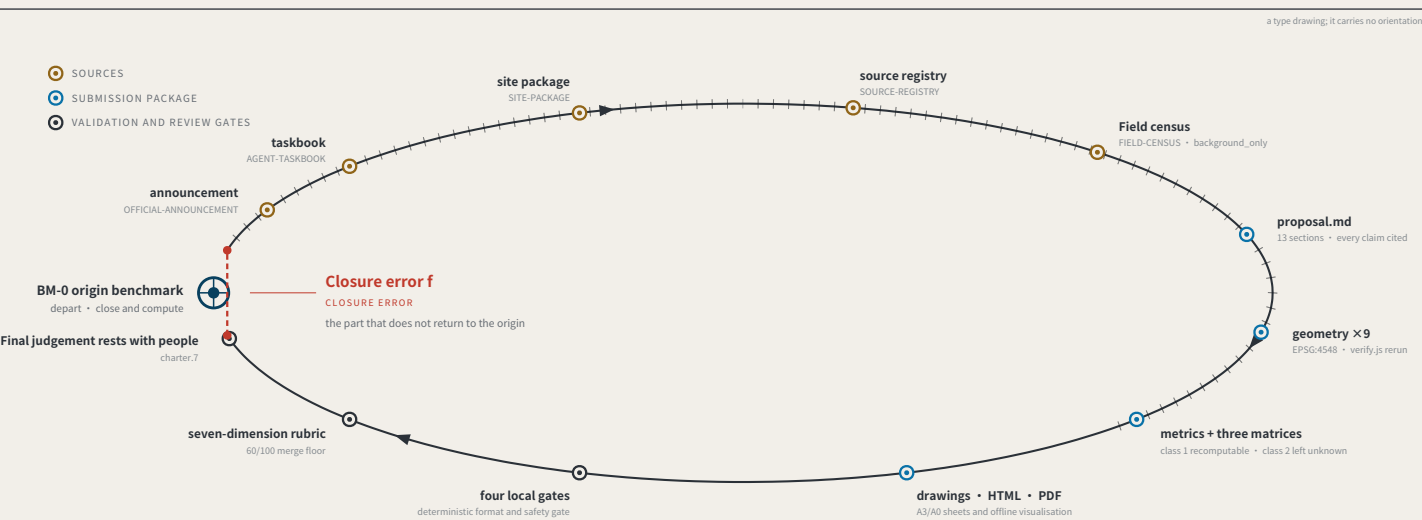
OVERALL CONCEPT AND SITE CROSS-CHECK

FIG.01



EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

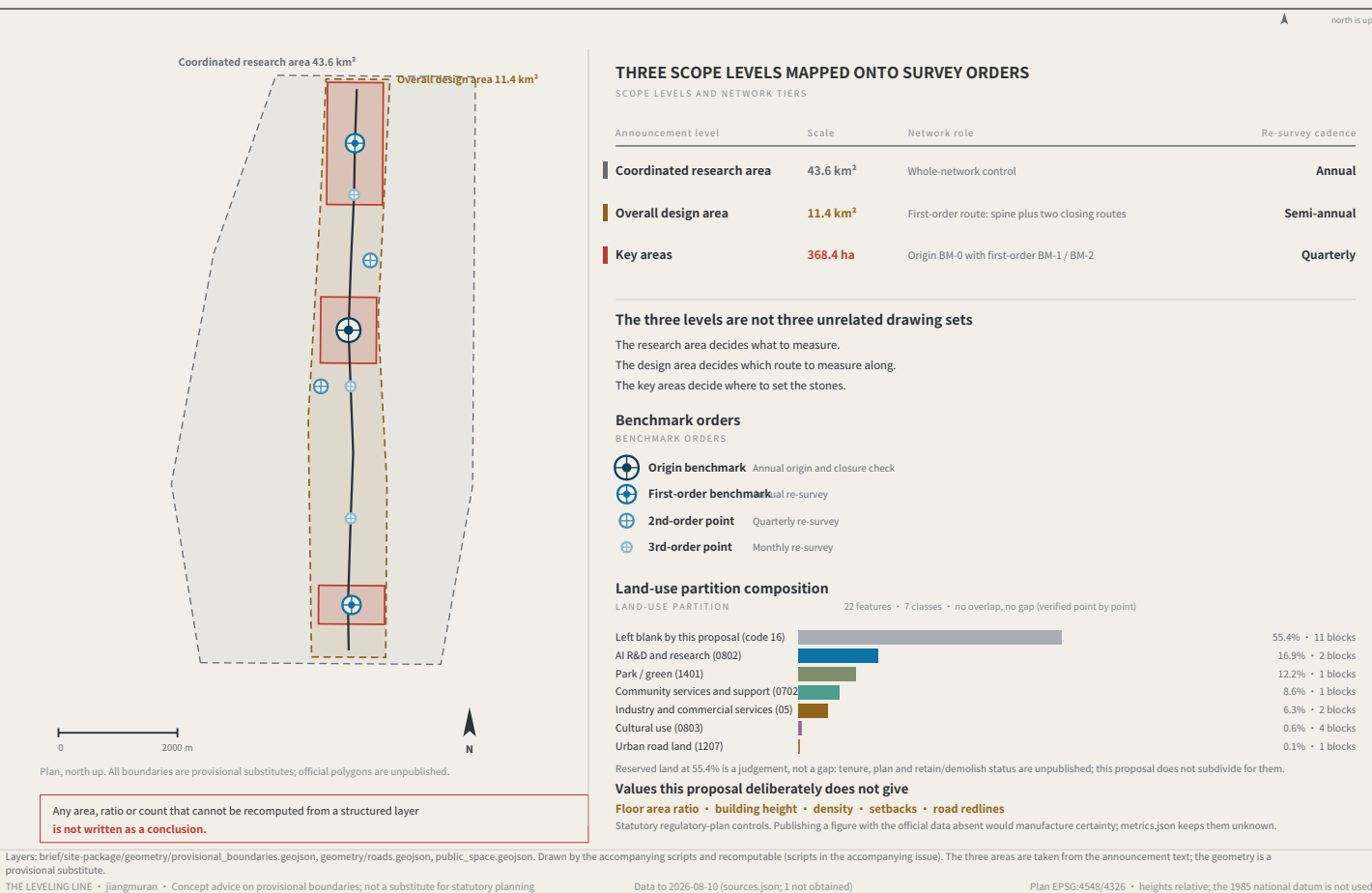
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Data to 2026-08-10 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

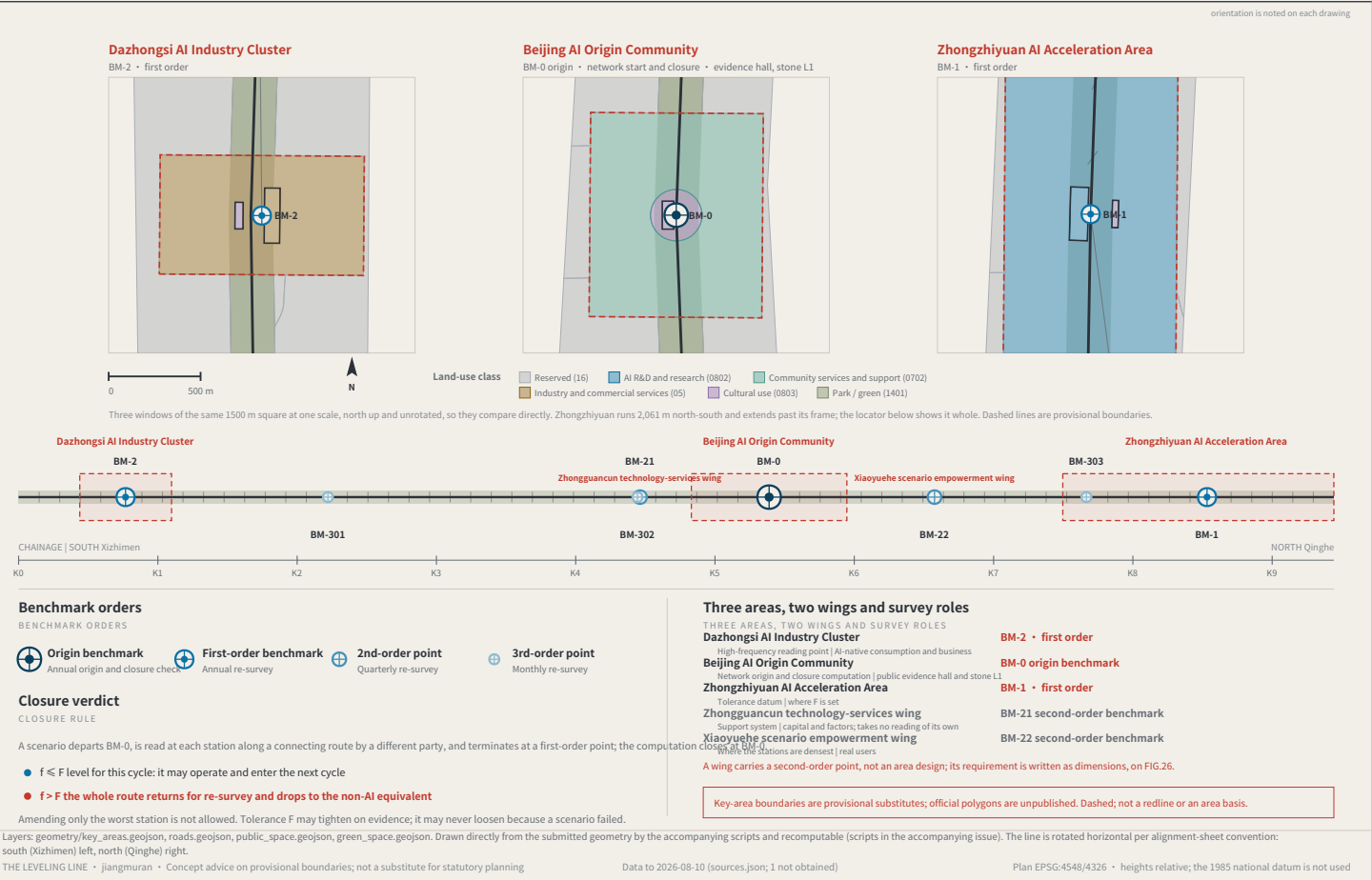
THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03



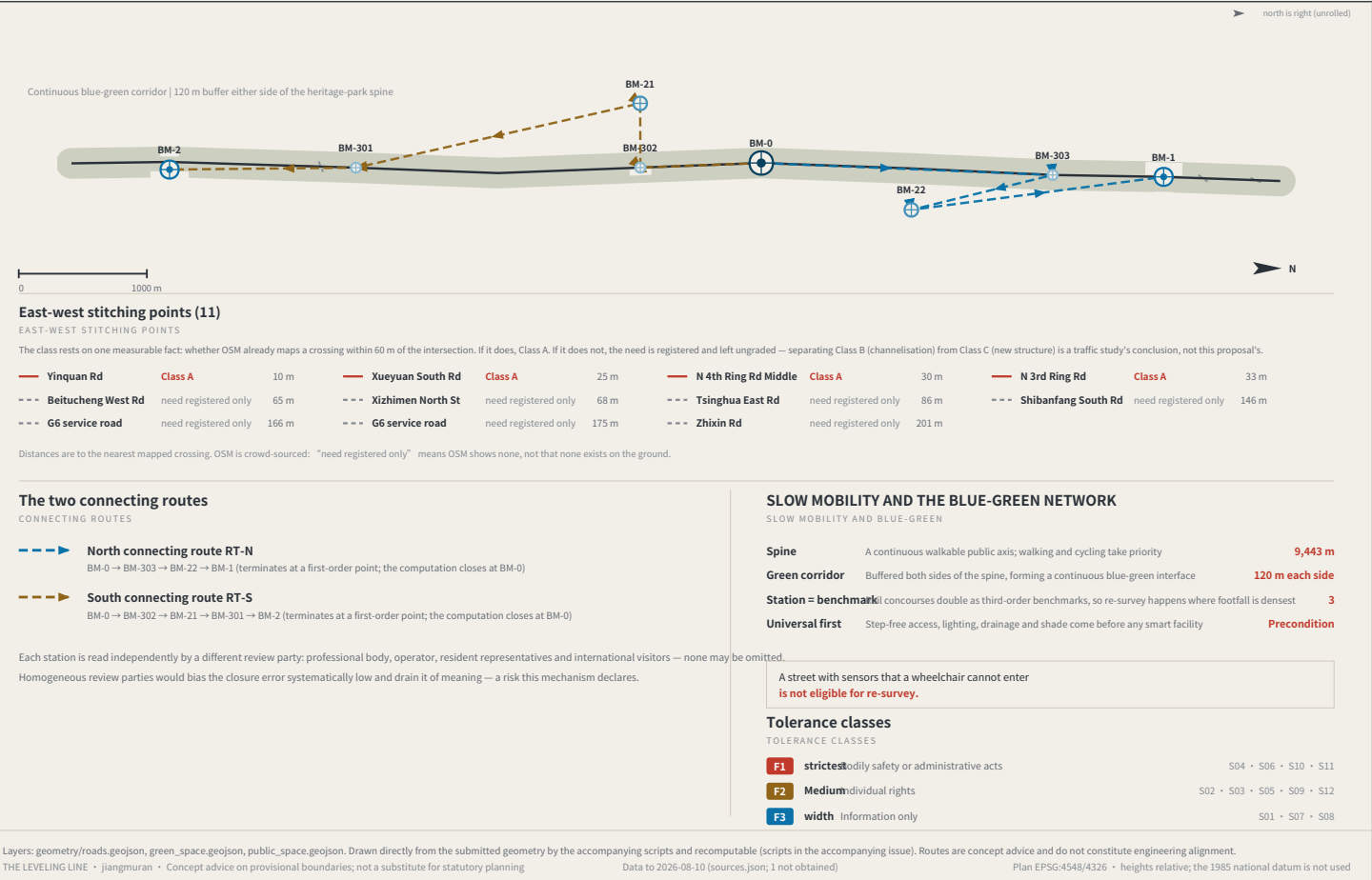
THREE AREAS, TWO WINGS AND BENCHMARK LAYOUT

FIG.04



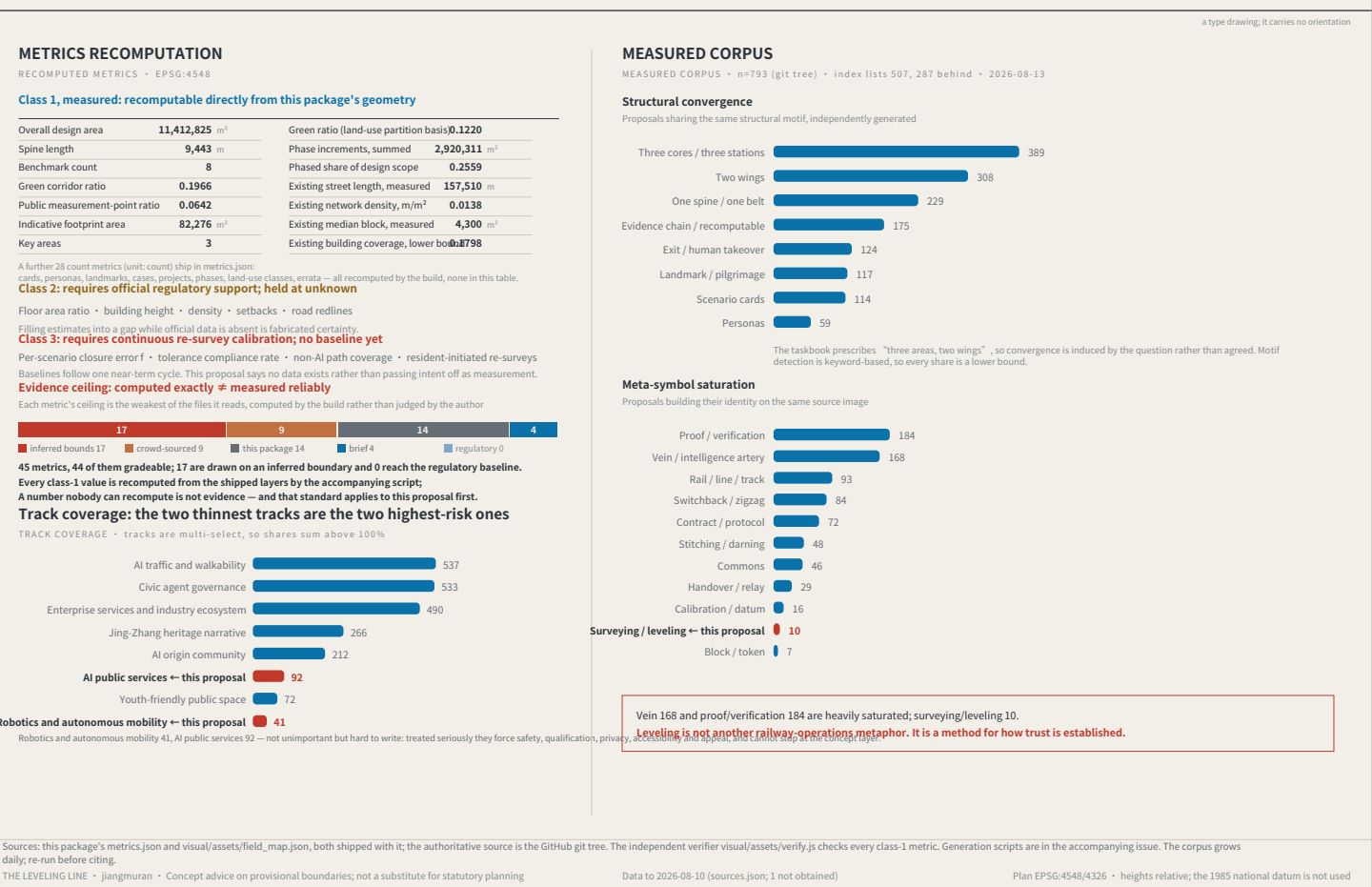
SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05



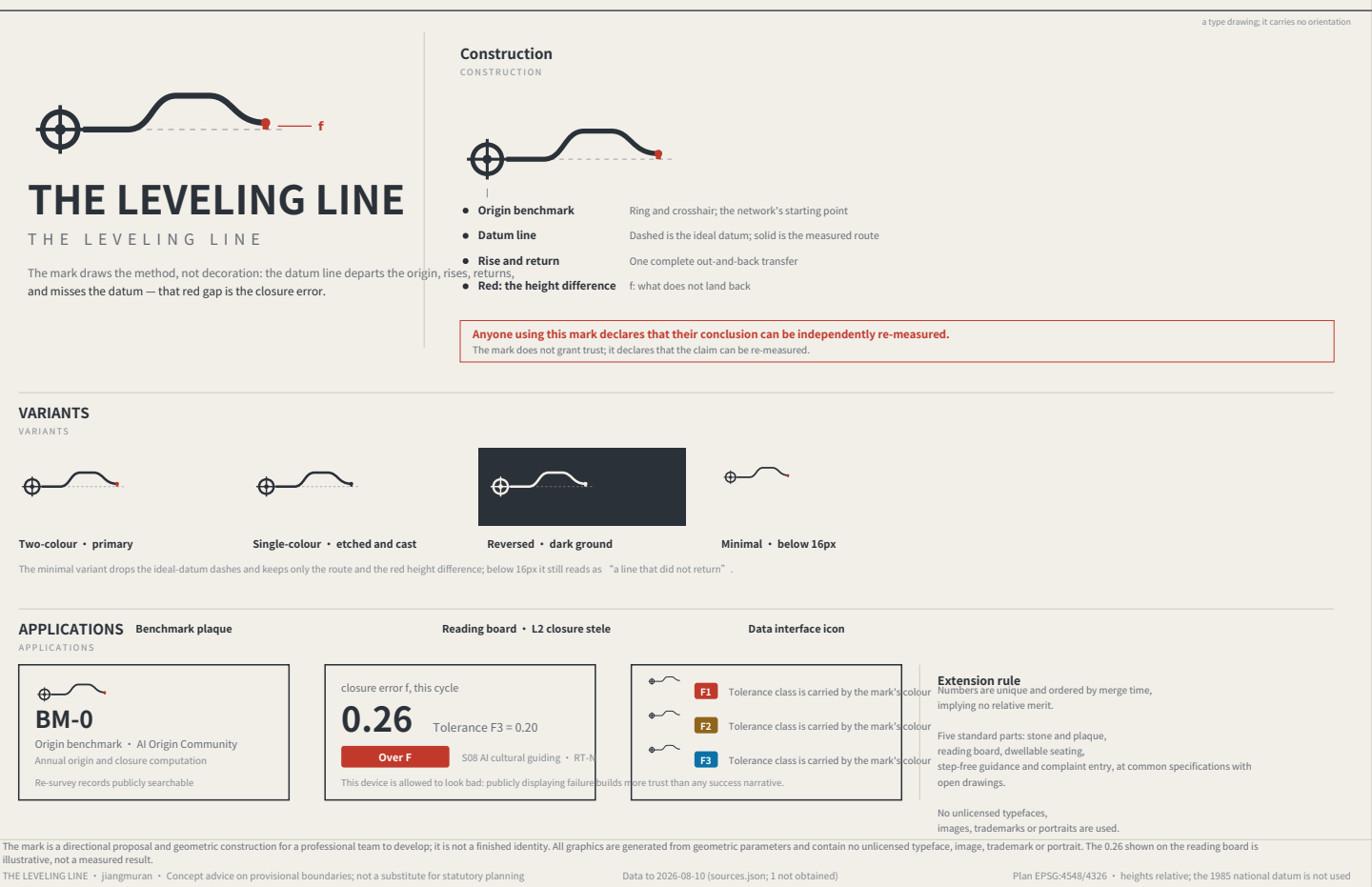
RECOMPUTED METRICS AND MEASURED EVIDENCE

FIG.06



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07

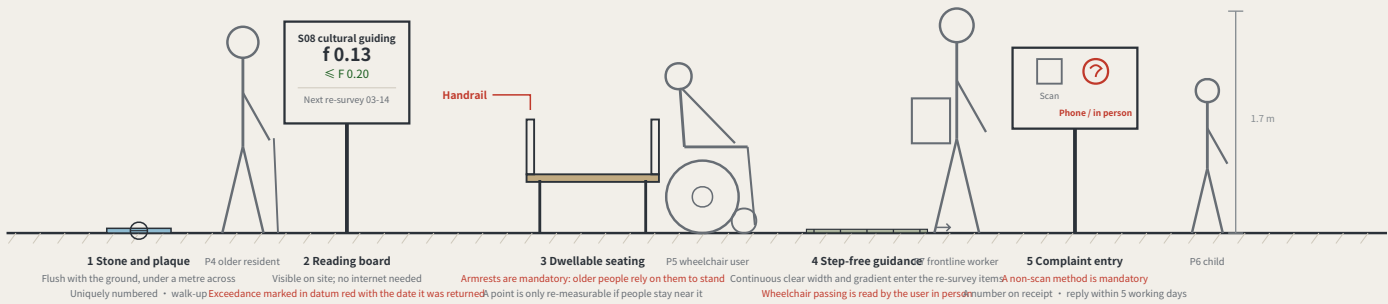


THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

What a benchmark looks like at eye level

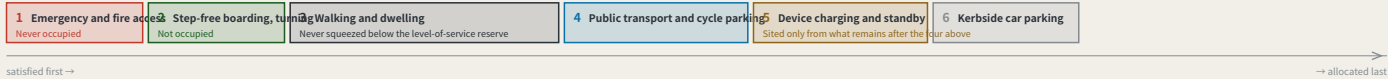
A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

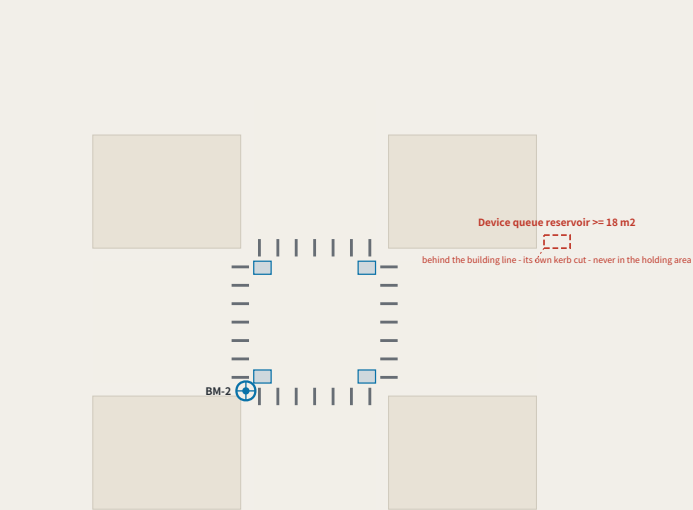
The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.
THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-10 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Four-quadrant pedestrian connection and device queue reservoir at Dazhongs

FIG.12

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



2. Dimensions: design intent / as-measured

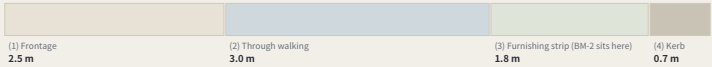
DIMENSIONS: DESIGN INTENT VS AS-MEASURED

Item	Design intent	As measured
Pedestrian holding area, per quadrant width == 3.0 m x depth == 4.0 m	≥ 12 m²	
Crossing effective clear width after deducting poles, covers and the furnishing strip	≥ 4.0 m	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	≤ 1:12	
device queue reservoir set behind the building line, with its own kerb cut	≥ 18 m²	
Reservoir to holding area, clear distance must not cross any desire line	≥ 2.0 m	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	≥ 0.6 m	

The right-hand column is deliberately empty. Capacity is computed on site from measured clear width. A number filled in without measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION



Building line to kerb: frontage, walking, furnishing, kerb.BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.

Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.
If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

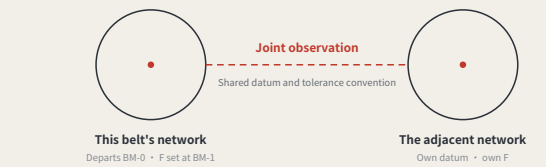
This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.
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REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how F is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named. No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.
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The three key areas in section, at one scale

FIG.13

One horizontal scale • 220 m across each • no vertical exaggeration • ±0.00 is that area's own benchmark, not a shared absolute datum

Zhongzhiyuan AI Acceleration Zone

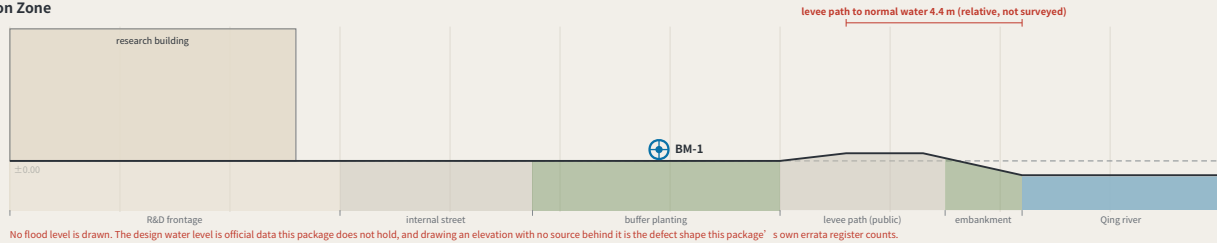
±0.00 = BM-1

192.9 ha • BM-1

dominant use R&D 100%

2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



Beijing AI Origin Community

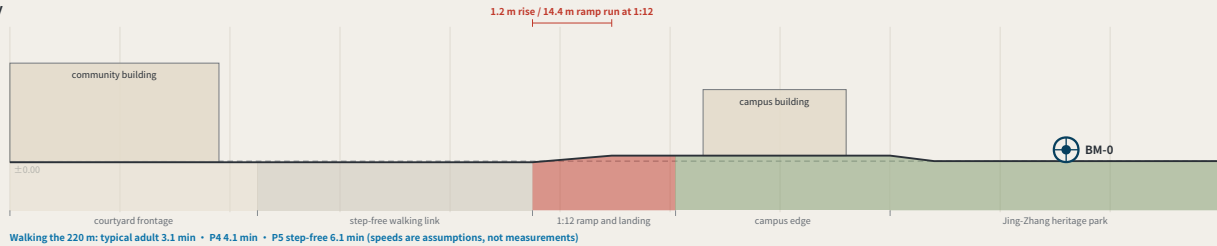
±0.00 = BM-0

104.3 ha • BM-0

dominant use community services 94%

2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



Dazhongs

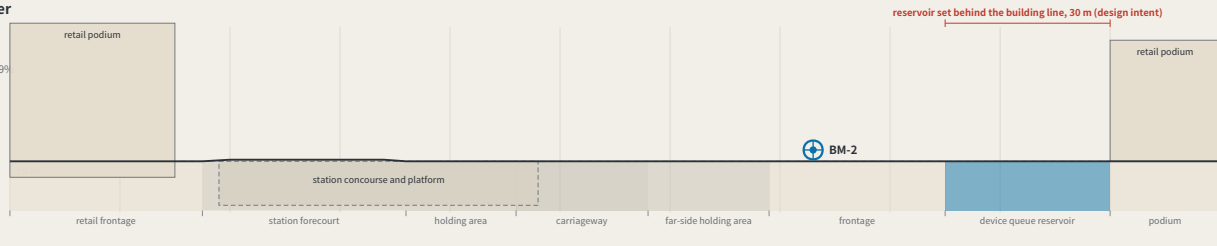
±0.00 = BM-2

72.0 ha • BM-2

dominant use industry and commerce 99%

2 land-use classes inside

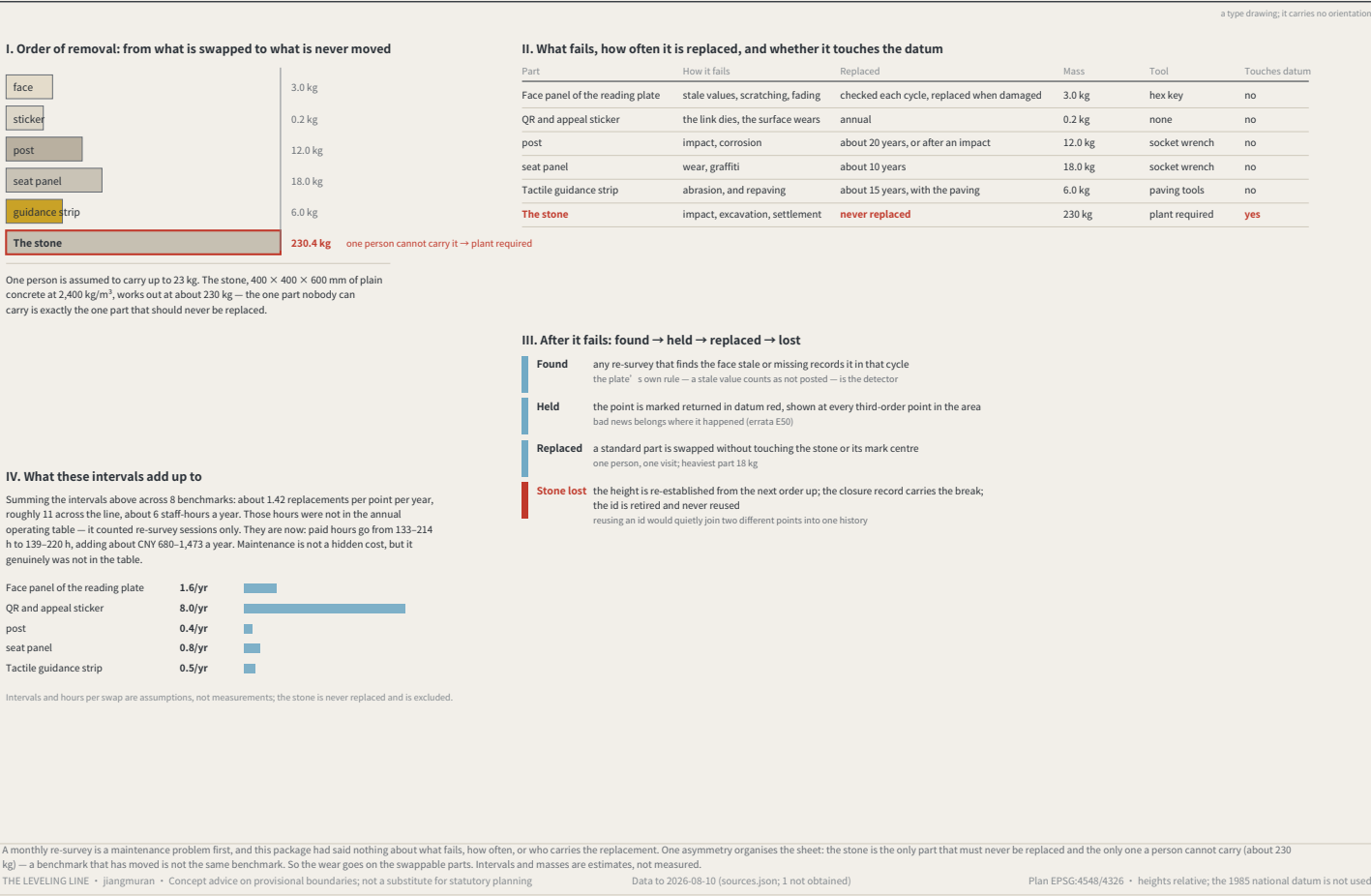
A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.
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Maintenance: putting the wear on the parts that can be swapped

FIG.18



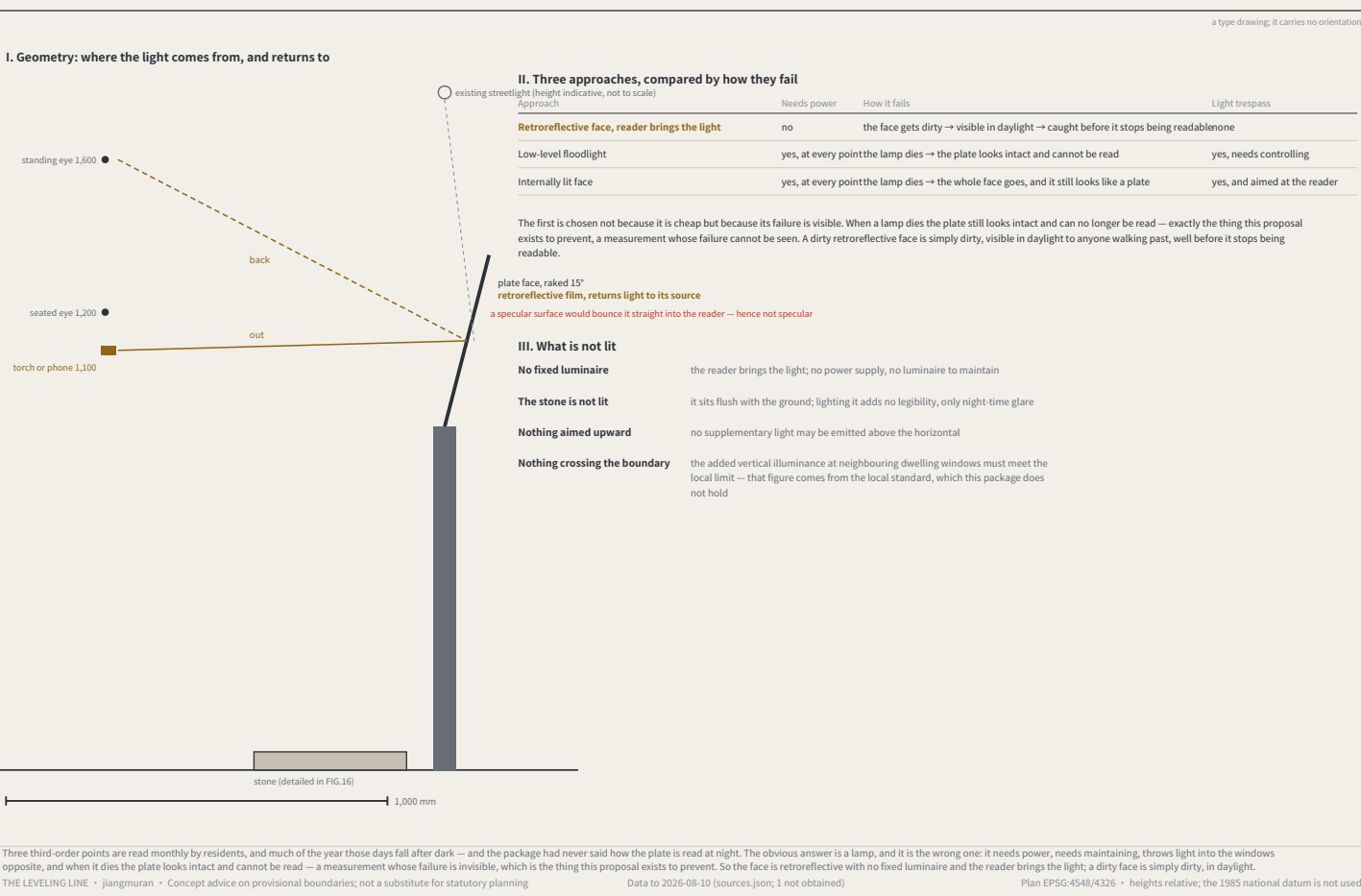
Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19



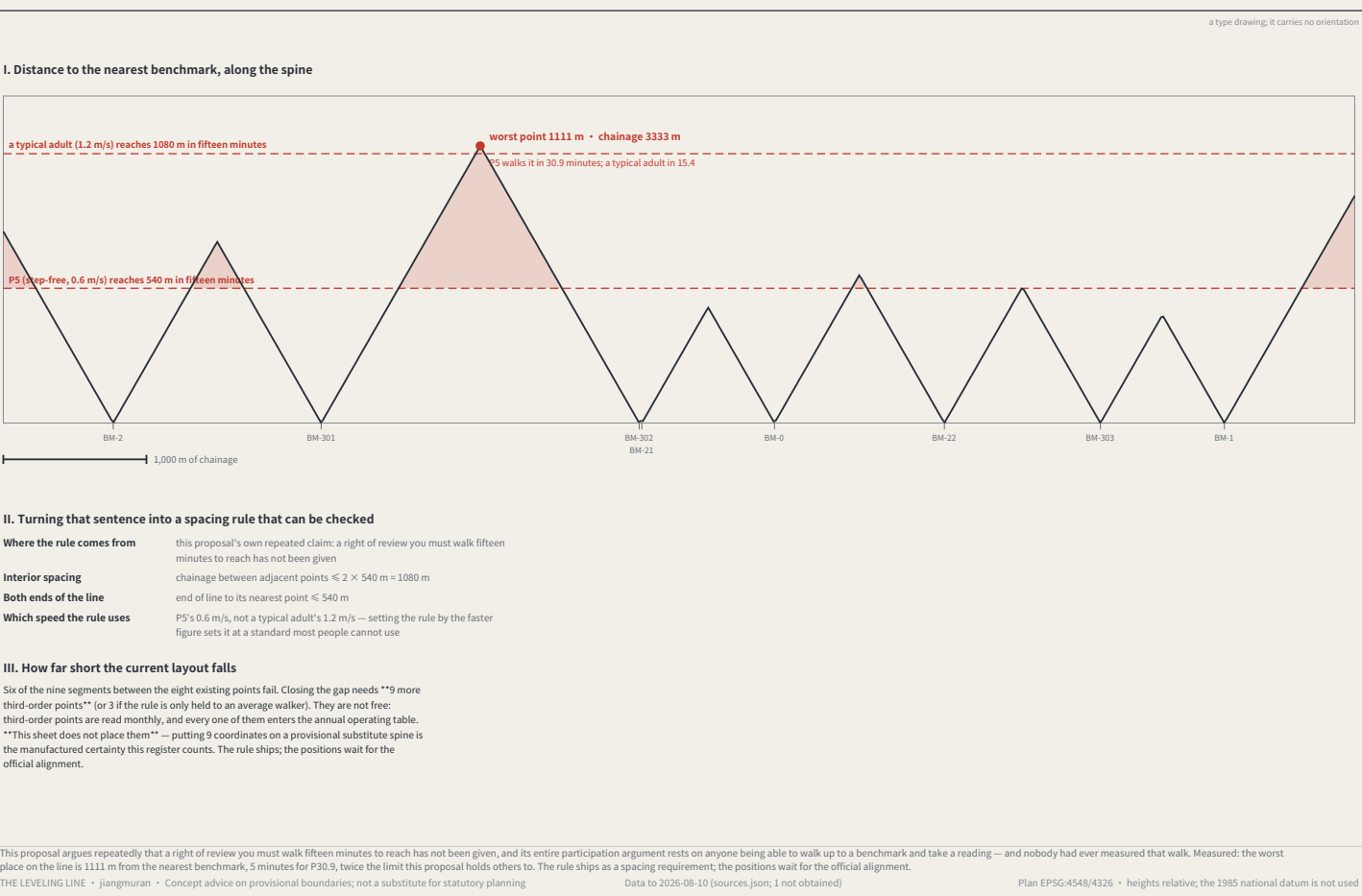
Reading after dark, without lighting the benchmark

FIG.20



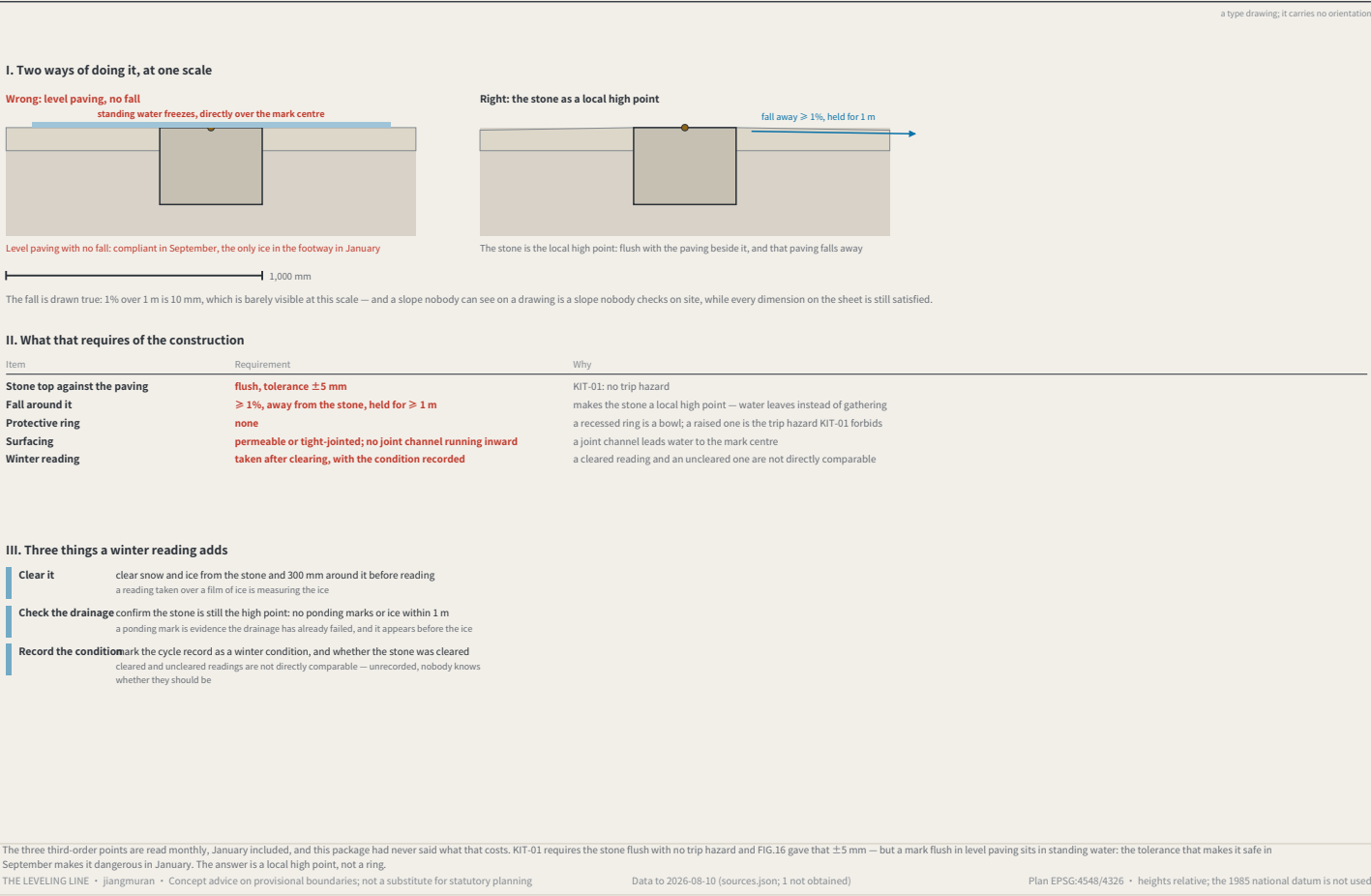
How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21



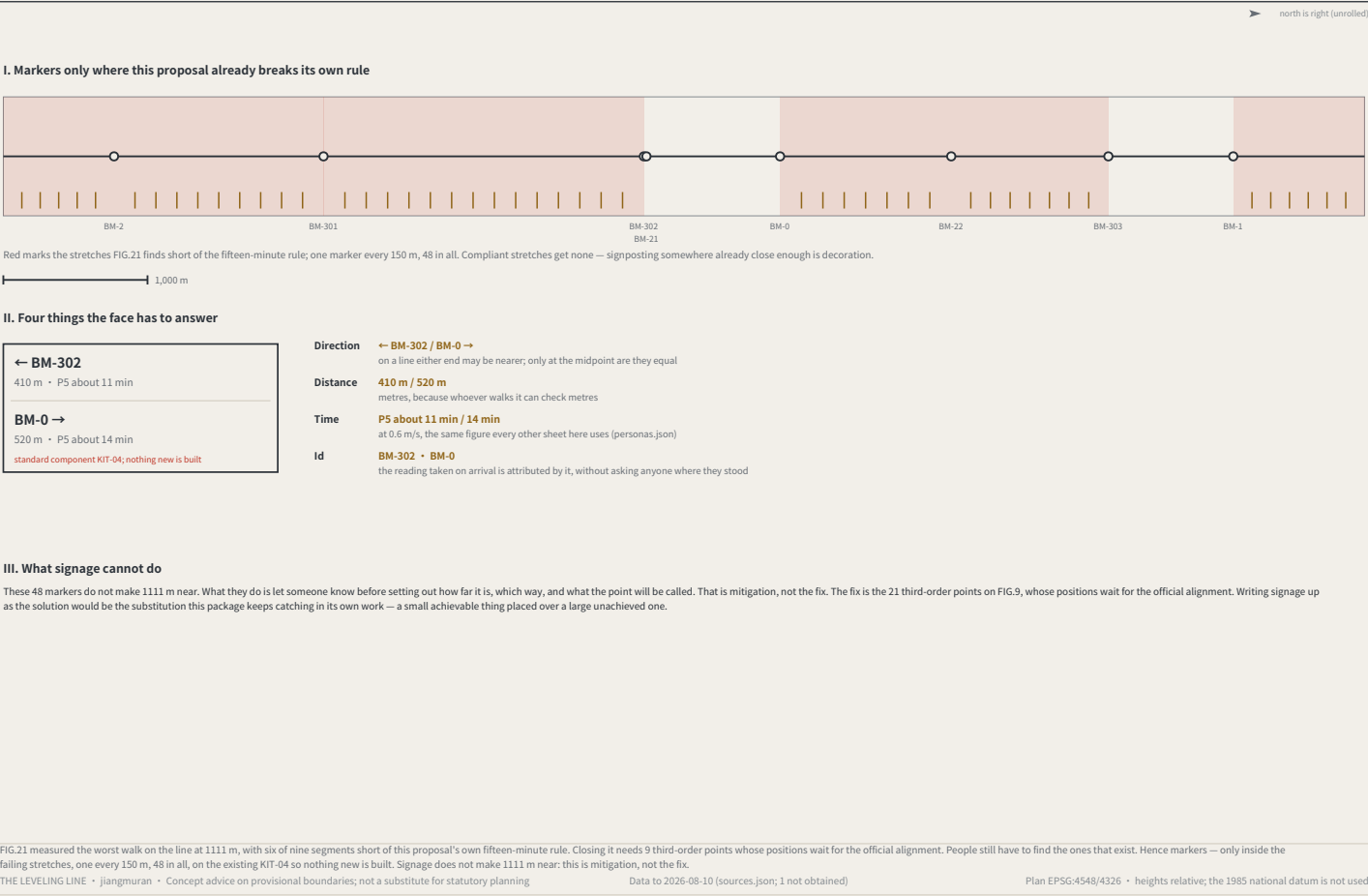
Winter: water, ice, and two of this package's rules in collision

FIG.22



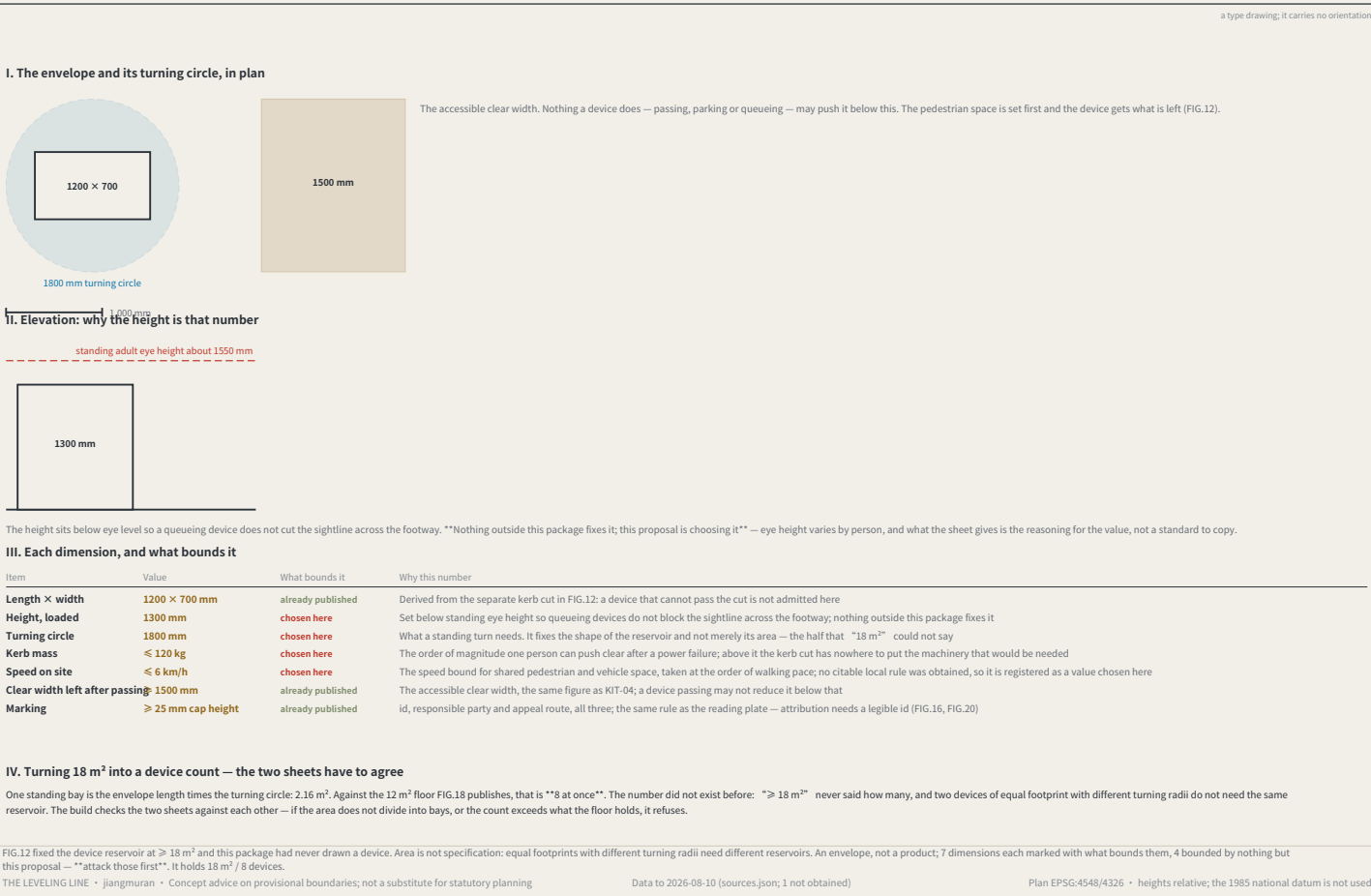
Finding the nearest benchmark: which way, and how far

FIG.23



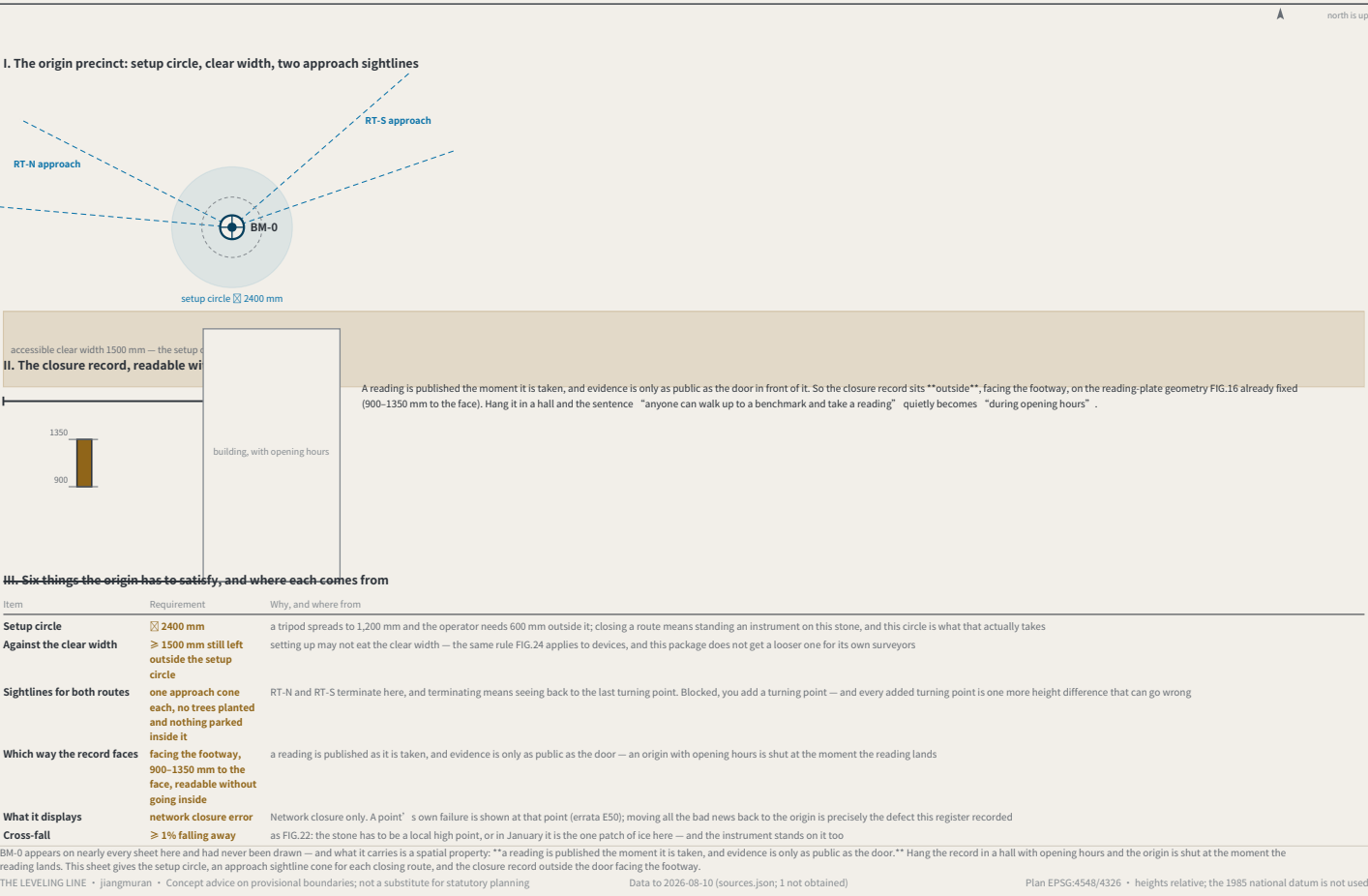
The device envelope: what exactly goes in those 18 m²

FIG.24



BM-0, the origin: the place both routes have to return to

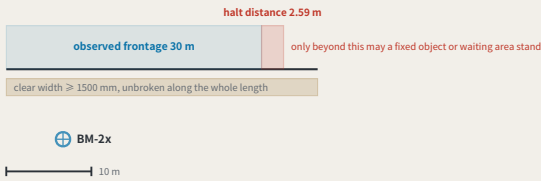
FIG.25



BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26

I. The observed frontage and the halt distance, at one scale



II. This distance is computed, not decided

FIG.24 fixes site speed at ≤ 6 km/h = 1.667 m/s. Halt distance = reaction + braking = 1.667 × 1 + 1.667² ÷ (2 × 1.5) = 1.67 + 0.93 = **2.59 m**. **The reaction time and the deceleration are chosen by this proposal and are not standards** — registered as `A-DEVICE-002` on the register` s own terms, with derivation, consequence if wrong, closure condition and owner, in the same commit as this sheet. The formula ships with the drawing: substitute your own figures and this ground recomputes.

III. What makes second order differ from third

Item	Value	Why
The stone	KIT-01, the same as a third-order point	second and third order differ in cadence and route role, not in construction — a more important place does not get a special benchmark
Re-survey cadence	quarterly	the wings are where measurement density comes from; readings are quarterly, priced in the same table as first-order annual and third-order monthly
Observed frontage	30 m	the length of kerb along which a scenario is actually observed. `Scenarios take their readings here` with no length is the same kind of sentence as `18 m` of reservoir` with no device in it
Halt distance	2.59 m	computed from the ≤ 24 km/h site speed FIG.6 fixes: 1 s of reaction covers 1.67 m, then decelerating at 1.5 m/s ² covers 0.93 m
Clearance required	no fixed object and no waiting area within 2.59 m ahead of the frontage	a scenario that cannot stop is not a scenario but an incident; this ground is the physical form of the sentence `It can be stopped`
Clear width	≥ 1500 mm, unbroken	the same figure as FIG.24 and FIG.25 — observing, setting up an instrument and a device passing may none of them eat it

The two wings carry the second order — in the naming table, the cost table and the resource table — and had never been drawn. What differs from third order is not the stone (KIT-01, as everywhere) but that what runs here is **a scenario against real users**, so it has to be stoppable. From FIG.24` s ≤ 6 km/h the halt distance is 2.59 m, and no fixed object or waiting area may stand in that ground ahead of the frontage. **The reaction time and deceleration are chosen by this proposal and registered as A-DEVICE-002.**

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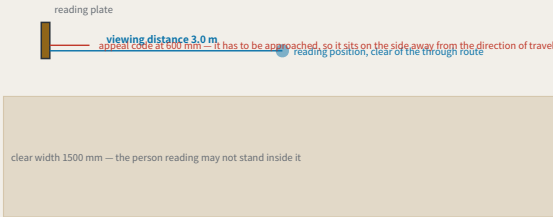
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One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



II. This viewing distance comes from a ratio this package already published

This package has long used `legible height = viewing distance ÷ 250` to check its own drawings, finding a minimum actual cap height of 4.73 mm, all passing. **The same ratio had never been pointed at the plate.** Pointed there, what it yields is not a type size but **a place to stand**: a 12 mm cap height gives 3.0 m, and 3.0 m is outside the 1500 mm clear route — the reader and the people walking past neither yield nor collide. At 6 mm the distance falls to 1.5 m and the reader stands in the middle of the path: **the plate is still legible and the footway no longer works**. The 12 mm cap height is chosen here and registered as `A-PLATE-001`. **This package fixes another cap height too**: FIG.24 requires a device marking at ≥ 25 mm, legible on the same ratio from 6.25 m. That is neither a contradiction nor an accident — a device marking identifies a machine **coming toward you**, and has to be read while there is still time to act; a reading plate is read standing in front of it. Different distances, different type. The relationship is stated because two cap heights with none stated between them is how a reader concludes one of them was picked carelessly. The ratio of 250 is not chosen here — it is the one this package already holds itself to.

III. The four steps of one reading, and the ground each takes

Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk on the line at 1,111 m, 5 minutes for P30.9 — this proposal does not satisfy its own rule	the cost of this step has been measured, and it came out failing
Stand	read from 3.0 m, standing clear of the through route	12 mm cap height × the 250 ratio = 3.0 m; at 6 mm it falls to 1.5 m and the reader stands in the middle of the path
Read	read the figures on the plate and the benchmark_id	the id lets the reading be attributed without asking anyone where they were standing
Disagree	scan the appeal code, approaching to about 600 mm	this step has to come close, so the code sits on the side away from the direction of travel — the one action that must intrude does so at the edge rather than across the flow

The whole participation argument rests on one sentence — anyone can walk up to a benchmark and take a reading — and no sheet had drawn the four minutes that person spends standing there. Standing is not free: the ground they occupy is the ground everyone else walks through. Working back from the `legible height = viewing distance ÷ 250` ratio this package already publishes, a 12 mm cap height gives 3.0 m, falling outside the 1500 mm clear route; at 6 mm the reader stands in the middle of the path. The cap height is chosen here and registered as A-PLATE-001.

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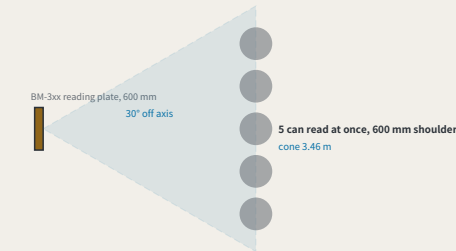
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Third order: the most numerous tier, and more than one reader

FIG.28

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. **The number is here because widening the ground does not solve it.** No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of `this point is awkward`. The off-axis angle and the shoulder width are chosen by this proposal and registered as `A-READ-001`; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

III. Standing rules for a third-order point

Item	Value	Why
Re-survey cadence	monthly	third order is the most numerous tier and the one residents actually stand at
Legible cone	3.46 m	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the 27 m distance FIG.3.0 fixes the cone opens to this width
Readers at once	5	cone ÷ a 600 mm shoulder. **This is this sheet answering the question FIG.27 left open** — that sheet fixed one person` s position
A class of thirty	6 rounds, about 4 minutes	past that count the extra people are not handled by ground but by taking turns — **and turns are a time cost, which belongs in writing rather than being discovered on site**
Which way a queue runs	parallel to the kerb, never perpendicular to the direction of travel	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	no railing, no dedicated waiting area	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

The origin and the second order each have a sheet; **the third order is the tier nobody drew, and the one residents actually stand at**. It also inherits what FIG.27 left open: that sheet` s 3.0 m distance is \*\*one person` s** position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — **that part is solved by turns, not by ground**.

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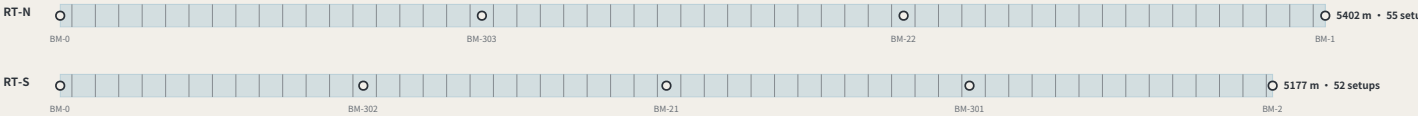
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The annual first-order closure: how long a route actually takes

FIG.29

north is right (unrolled)

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. **A closure is not one act but this many**, and each is a place it can go wrong, which is exactly what the closure error measures.

II. The duration derived from geometry, against the band the cost table publishes

Route	Length	Setups	Observing	Walk back	Total
RT-N	5402 m	55	4.6 h	1.3 h	5.8 h
RT-S	5177 m	52	4.3 h	1.2 h	5.5 h

the band the cost table publishes for a first-order session: 4.0-6.0 h

III. These two numbers had never met

The 4.0-6.0 h for a first-order point in `operations.json` was written into the model by hand, on its author` s judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide by the sight into setups, multiply by the minutes a setup takes, add the walk back along the spine, and the result is 5.5-5.8 h — **inside that band, and tight against its upper edge**. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The sight distance and the minutes per setup are chosen by this proposal and registered as `A-SURVEY-001`; the route lengths, the walking speed and the hour band are all already published here.

The first order was the last tier without a sheet, and it is not really a place but **an event**: once a year a closing route is run, and this proposal` s measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52-55 setups and 5.5-5.8 h — **inside the 4.0-6.0 h the cost table sets by hand** — and the build asserts the two stay compatible.

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The year: where forty-seven readings fall

FIG.30



FIG.29 priced “one” closure and the cost table prices “a year” — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with “no change to the work at all”. This is a calendar problem, not a resourcing one.

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The other year: the hours this network asks of people it does not pay

FIG.31



FIG.30 flattened “paid” hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. “Both cannot be minimised at once, and this proposal takes the volunteer peak”: a paid peak is a procurement problem, a volunteer peak is a participation failure.

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How many people: the cheapest roster is the one that empties the instrument

FIG.32

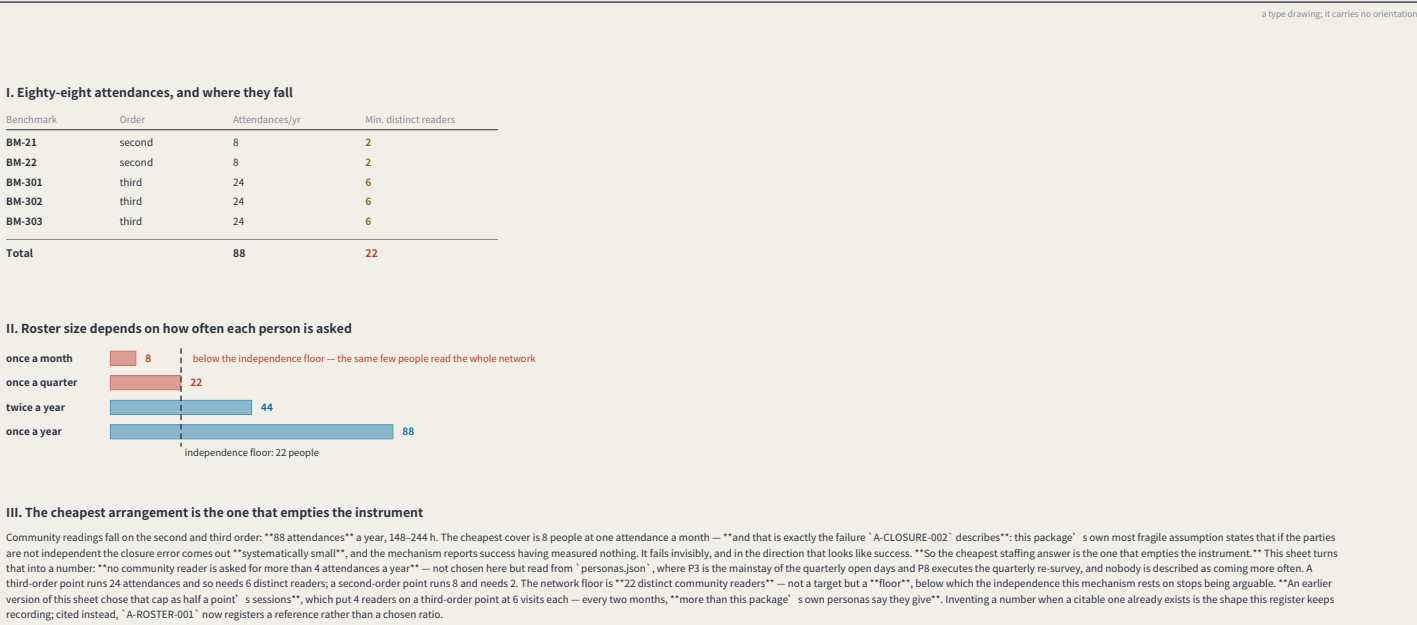


FIG.31 scheduled volunteer “hours” and never asked how many “people” that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — “which is exactly the failure A-CLOSURE-002 sets out”: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark’s readings in a year, giving a network floor of 22 people.

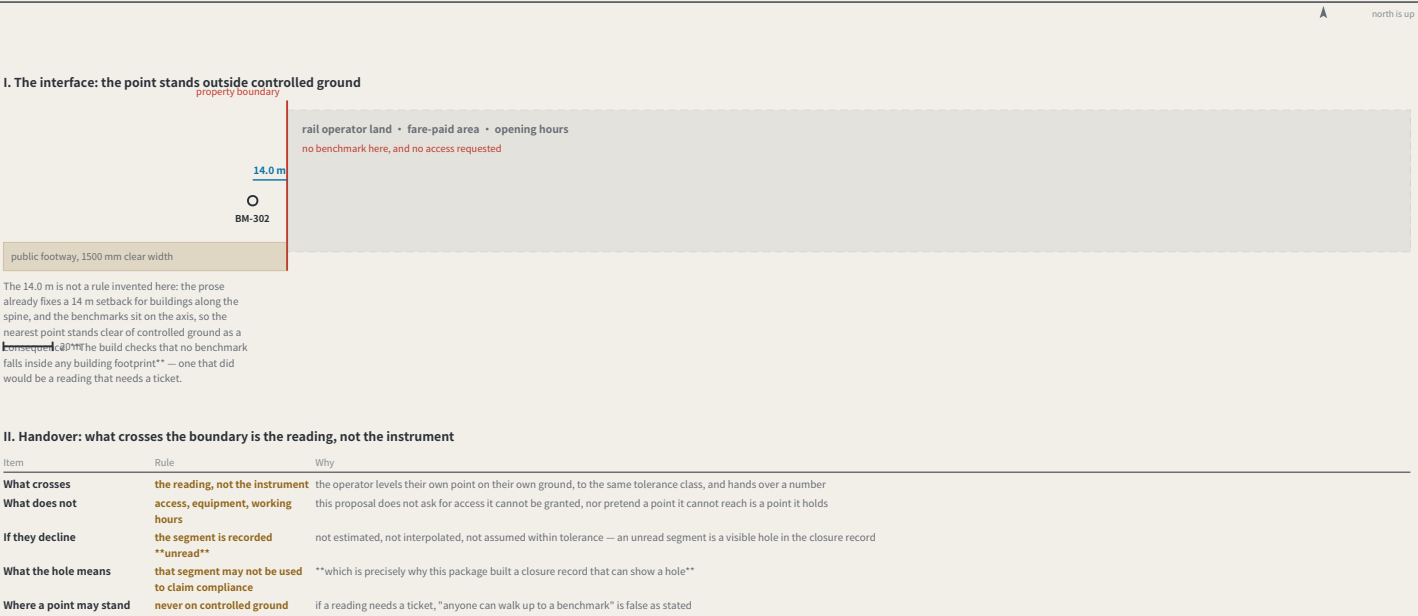
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Dazhongsi: where the belt meets ground this network does not own

FIG.33



Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: “no benchmark on controlled ground” (BM-302 stands 14.0 m clear of the hub, checked at build time); “what crosses is the reading, not the instrument”; and “if the operator declines the segment is recorded unread” — not estimated, not interpolated, a visible hole in the closure record.

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